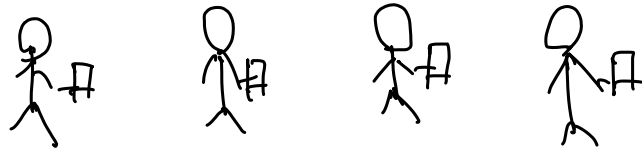


last: Direct proof

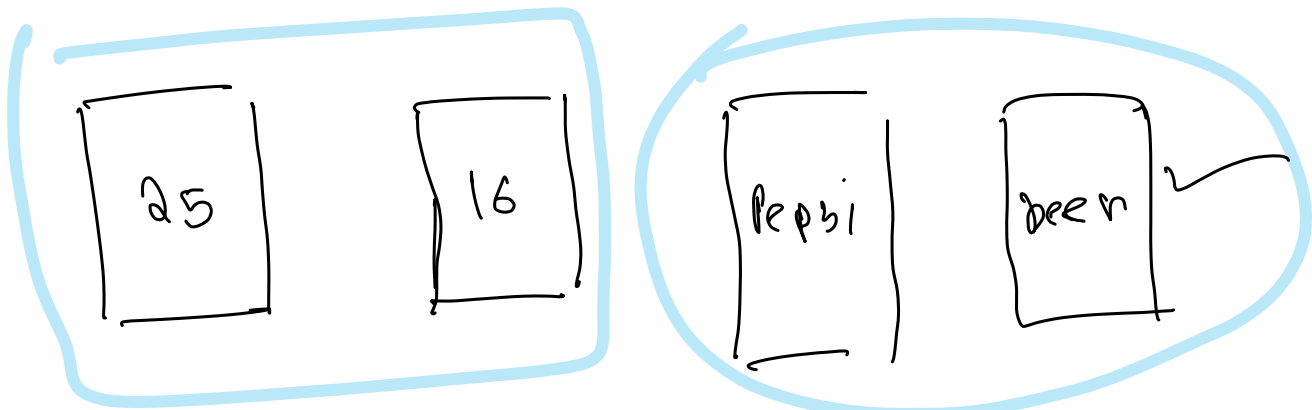
if P then Q $(P \Rightarrow Q)$

Contrapositive

say: there are 4 people everyone is holding a drink



- if a person is taking Alcohol, then he/she is over 21
if P then Q
- if we know the person is under 21, the statement will become false if they are taking Alcohol
- if we knew alcohol is being consumed, then the 'implication statement' will be false if Q is false (Q: he/she is over 21)



for these two only way the statement can be false
if the card under 16 reads beer/alcohol

✓ Alcohol $\Rightarrow$ age $>$ 21 ✓

- if someone is drinking then he/she is over 21
- if someone is not 21 then, he/she is not drinking

1st statement: $P \Rightarrow Q$, if P then Q

2nd statement: $\neg Q \Rightarrow \neg P$, if not Q then not P

Quick question: $P \Rightarrow Q \quad \neg P \wedge Q$
 $\swarrow \quad \searrow$
 $\neg Q \Rightarrow \neg P$

Proof structure: $P \Rightarrow Q$

Proof: Assume Q is false,

$\neg Q$ is true

⋮
 $\neg P$ is true

Proposition: if P then Q

$x \in \mathbb{Z}$, if $x^2 - 6x + 5$ is even then x is odd
 $\quad \quad \quad P \quad \quad \quad Q$

Direct proof: $x^2 - 6x + 5$ is even true,

$$x^2 - 6x + 5 = 2k$$

$$\Rightarrow x^2 - 6x + (5 - 2k) = 0$$

Contrapositive, $(\sim Q)$
So, x is not odd, x is even,

$$\therefore x = 2k \quad [k \in \mathbb{Z}]$$

Now, in $x^2 - 6x + 5$

$$= (2k)^2 - 6 \cdot 2k + 5$$

$$= 4k^2 - (2k + 4 + 1)$$

$$= 4(k^2 - 3k + 1) + 1$$

$$= 2 \cdot 2(k^2 - 3k + 1) + 1$$

We know, $2(k^2 - 3k + 1)$ is $\mathbb{Z}$

We take, $2(k^2 - 3k + 1) = 2k_1 \quad [k_1 \in \mathbb{Z}]$

$$\therefore x^2 - 6x + 5 = 2k_1 + 1 \quad [k_1 \in \mathbb{Z}]$$

for x is even

$\therefore x^2 - 6x + 5$ is odd,

$\therefore x^2 - 6x + 5$ is not even,
 $(\sim P)$

$$(\sim Q) \Rightarrow (\sim P) \quad \therefore P \Rightarrow Q$$

if $x^2 - 6x + 5$ is even, then x is odd [shown]

• $a, b \in \mathbb{Z}$, if $a^2(b^2 - 2b)$ is odd then a and b both are odd,

Contrapositive:

if a and b both are not odd,

from De Morgan,

either a is even or b is even

Case : 1, a : even, b : odd

$$a = 2k_1 \quad b = 2k_2 + 1 \quad [k_1, k_2 \in \mathbb{Z}]$$

Now,

$$(2k_1)^2 \{ (2k_2 + 1)^2 - 2 \cdot (2k_2 + 1) \}$$

$$= 2 \cdot \underbrace{2k_1^2 (2k_2 + 1) (2k_2 - 1)}_{\text{integer} = k_3} \text{ is even} \quad \text{how?}$$

hence, for,

a : even, b : odd,

$a^2(b^2 - 2b)$ is even

$\therefore a^2(b^2 - 2b)$ is not odd,

Case : 2:

a : odd, b : even,

$$a = 2k_1 + 1, \quad b = 2k_2$$

$$[k_1, k_2 \in \mathbb{Z}]$$

Now,

$$(2k_1+1)^v ((2k_2)^v - 2k_2)$$

$$= 2k_2 (2k_1+1)^v (2k_2-1) \sim \text{int}$$

$\therefore$ for a is odd and b is even

$a^v(b^v-2b)$ is not odd,

Case: 3 $a = \text{even}$, $b = \text{even}$

$$\therefore a = 2k_1, \quad b = 2k_2 \quad [k_1, k_2 \in \mathbb{Z}]$$

$$\begin{aligned} \therefore a^v(b^v-2b) &= (2k_1)^v ((2k_2)^v - 2 \cdot 2k_2) \\ &= 2 \cdot 2k_1^v (4k_2^v - 4k_2) \sim \text{int} \end{aligned}$$

for $a = \text{even}$, $b = \text{even}$

$a^v(b^v-2b)$ is not odd,

for every case, $\neg Q \Rightarrow \neg P$

hence $P \Rightarrow Q$

$a^v(b^v-2b)$ is odd $\Rightarrow$ a and b both are odd,

• suppose $x, y \in \mathbb{R}$, if $\frac{y^3 + yx^2}{p} \leq \frac{x^3 + xy^2}{p}$ then $y \leq x$,

Q

Proof: suppose, $y \leq x$ is false,

then, $y > x$

$$\rightarrow y - x > 0$$

since, $x, y \in \mathbb{R}$, $x^2 + y^2 \geq 0$

$x^2 + y^2 > 0$ since, $y > x$

multiplying $(y-x)$ on both sides,

$$(y-x) \cdot (x^2 + y^2) > 0$$

$$\rightarrow x^2 y + y^3 - x^3 - x y^2 > 0$$

$$\rightarrow x^2 y + y^3 > x^3 + x y^2$$

$\therefore$ if $y > x$ then $x^2 y + y^3 > x^3 + x y^2$

if not $y \leq x$ then $x^2 y + y^3 \leq x^3 + x y^2$ is not true

• let, $a, b \in \mathbb{Z}$, let p be a prime, if $p \nmid ab$ then $p \nmid a$ and $p \nmid b$.

R

if $\forall R$ then $\forall S$

lemma: (Cummings book)

(2.17) let a, b and c be integers and let p be a prime,

(iii) if $p \mid bc$ then $p \mid b$ or $p \mid c$ or both

$\left. \begin{array}{l} \text{if } R \text{ then } S \\ \text{if } \sim R \text{ then } \sim S \end{array} \right\} \text{(not contrapositive)}$

Proof:

$$p \nmid ab, \quad ab = m \cdot p + k \quad k \neq 0$$
$$\downarrow$$
$$(m_1 p + k_1) \cdot (m_2 p + k_2)$$

Contrapositive:

$p \nmid a$ & $p \nmid b$ is not true,

$p \mid a$ or $p \mid b$

without the loss of generality,
 $p \mid a$ by the defn, divisibility,

$$a = k_1 p \quad [k_1 \in \mathbb{Z}]$$

$$\rightarrow a \cdot b = (k_1 p) b$$

$$\rightarrow a \cdot b = p \cdot (k_1 b) \quad [\text{since } k_1 \in \mathbb{Z} \text{ and } b \in \mathbb{Z}]$$
$$\therefore k_1 b \in \mathbb{Z}$$

so, from the defn of divisibility,

$$p \mid a \cdot b, \quad \text{hence } \sim Q \Rightarrow \sim P$$
$$\therefore P \Rightarrow Q$$

- if a is an odd integer,
then, $x^2 + x - a^2$ has no integer solⁿs.